

Portfolio README Generator

Automation Name

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What It Does

This automation generates GitHub-ready README files for completed projects and labs. Feed it a brief summary or raw code from a C++ tool or cybersecurity lab, and it outputs a professional, consistently formatted README that matches your portfolio style. No more manual formatting or hunting for old templates. Every project gets recruiter-grade documentation instantly.

Implementation Guide

Step 1: Choose Your Tool

Set this up as a Claude Project or a Custom GPT. Both allow persistent instructions and uploaded reference materials.

Step 2: Gather Your Context Materials

Before building, collect these items:

- One or two example READMEs from your existing GitHub portfolio that match your preferred style.
- A short document with your bio snippet, your core tech stack, and any repeated phrasing you use across projects.
- A sample project or lab summary so you can test the output immediately.

Step 3: Write Your System Instructions

Create the custom instructions for your tool. Include:

- Role: You are a documentation assistant specialized in creating GitHub README files for technical projects.
- Formatting requirements: Specify your preferred structure. Example sections include Project Overview, Features, Installation, Usage, Tech Stack, and Key Takeaways.
- Tone and style: Describe whether you want it concise or detailed, technical or accessible.
- Output rules: Instruct the tool to use Markdown, include code block placeholders, and format headings consistently.

Step 4: Upload Reference Files

Upload your example READMEs and tech stack document to the project or custom GPT. This gives the tool persistent access to your style and preferred language.

Step 5: Test With a Real Example

Drop in a quick summary of a recently completed project or lab. Review the generated README. Check formatting consistency, tone, and completeness. Iterate on your instructions if needed.

Step 6: Use It Weekly

Each time you finish a project or lab, open the tool, paste in your summary or code snippets, and generate the README. Copy the output directly into your GitHub repository.

Before You Build: Verify Your Tool Can Do This

- Persistent memory across sessions: Confirm your chosen tool can store custom instructions and uploaded files permanently. Claude Projects and Custom GPTs both support this. Standard ChatGPT conversations do not.
- File upload capability: You need to be able to upload your example READMEs and tech stack reference. Both Claude Projects and Custom GPTs support file uploads.
- Markdown formatting: Ensure the tool can output clean Markdown with headings, code blocks, and lists. All major AI tools handle this well, but test with one example to confirm output quality.

What This Tool Cannot Do Alone

This automation will not automatically commit files to GitHub or scrape data from your codebase. You will still need to manually copy the generated README into your repository and push it. The automation handles the writing and formatting, not the Git workflow.

Starter Prompt

You are a technical documentation assistant specialized in creating professional, GitHub-ready README files for software projects and cybersecurity labs.

Your job is to take a brief project summary, code snippets, or lab descriptions and output a complete, polished README.md file that follows a consistent structure and tone.

Output Structure:

Every README you generate should include these sections in order:

1. Project Title - Clear, descriptive name
2. Overview - One or two sentences summarizing what the project does
3. Features - Bullet list of key functionality
4. Tech Stack - Technologies, languages, and tools used
5. Installation - Step-by-step setup instructions with code blocks
6. Usage - Example commands or screenshots showing how to use the project
7. Key Takeaways - What was learned or what makes this project noteworthy

Formatting Rules:

- Use proper Markdown with heading levels, code blocks, and bullet lists
- Keep code blocks clearly labeled with language tags
- Use concise, professional language
- Maintain a technical but accessible tone
- Include placeholders for screenshots where appropriate

When I provide you with a project summary, code, or lab write-up, generate a complete README following this structure. If any section is unclear from my input, make reasonable assumptions based on typical project patterns and note what I should verify or expand.